Strategic Integration for Gappy AI Hackathon: NexusDesk and Class Notes Pipeline

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EXECUTIVE SUMMARY

This report outlines the strategic battle plan for the Gappy AI Hackathon, focusing on integrating two powerful pre-existing tools: NexusDesk and the Class Notes Pipeline. NexusDesk offers a comprehensive student dashboard with attendance tracking, timetable ingestion, and project management features, while the Class Notes Pipeline provides robust audio recording, compression, and AI transcription capabilities. The core strategy involves leveraging the Lemma SDK to seamlessly merge these components, creating an intelligent system that transforms raw academic data into actionable insights displayed on the NexusDesk frontend. This integration provides a significant competitive advantage, aiming for a high probability of success in the hackathon.

Existing Foundations: NexusDesk and Class Notes Pipeline

The team enters the Gappy AI Hackathon with a substantial advantage, possessing two highly functional, pre-built codebases poised for integration. This foundational strength significantly reduces development time and allows for a focus on sophisticated integration. The first component, NexusDesk, is a custom Neo-Brutalist dashboard designed specifically for Sophomore ECE students at NITK. It offers a comprehensive suite of features essential for academic management, including an interactive hourly vertical timeline for class scheduling and attendance toggles, an AI-powered timetable/calendar ingestion dropzone utilizing client-side pdf.js and multimodal Vision OCR, and ECE course attendance gauges complete with risk calculators. Additionally, NexusDesk incorporates a robust Kanban board categorized for Academics, Hardware Dev, Routine, and Personal tasks, alongside a dedicated ECE Hardware Project Tracker for managing breadboarding, simulations, and parts checklists.

The second key asset is the Class Notes Pipeline, a powerful command-line tool focused on streamlined audio processing and transcription. This pipeline is adept at recording system audio from sources like Zoom or other online streams, as well as raw microphone output, ensuring comprehensive capture of lecture content. A critical feature is its ability to compress large raw WAV files into much smaller MP3s on-the-fly, optimizing storage and transfer. Furthermore, it leverages advanced AI models for transcription, offering flexibility between cloud-based Gemini models and efficient offline Whisper models. The pipeline then intelligently formats these raw transcripts into structured Word documents with logical sections, ready for further processing and analysis.

- Two fully functional, pre-built systems (NexusDesk & Class Notes Pipeline) provide a strong head start.
- NexusDesk offers comprehensive academic management, timetable ingestion, and project tracking.
- Class Notes Pipeline specializes in audio capture, compression, and AI-powered transcription.

The Integration Imperative: Leveraging Lemma SDK

The central objective for the hackathon period is the seamless integration of NexusDesk and the Class Notes Pipeline, orchestrated through the Lemma SDK infrastructure layer. This integration will transform disparate tools into a unified, intelligent system capable of autonomously managing academic workflows. The architecture envisions audio and PDF inputs feeding into their respective pipelines: the Class Notes Pipeline for audio, leading to a Transcription Agent, and the NexusDesk dropzone for PDFs, processed by a Syllabus Parsing Agent. Both agents will populate centralized document and datastores.

The Lemma SDK acts as the brain, with a Learning & Execution Agent pulling information from the document store (raw transcripts) and structured datastores (courses/tasks) via semantic search and retrieval mechanisms. This agent will then triage and allocate information to update various components of the NexusDesk dashboard. Specific build steps outline the migration of the existing shell script-based recording and compression pipeline into Lemma SDK Workflows. This transition is crucial for orchestrating audio capture, compression, transcription, and further data processing within a cohesive, scalable framework. While the provided text cuts off, the implicit goal is to ensure the entire lifecycle from audio/syllabus ingestion to dashboard updates is managed by the Lemma SDK.

- The core task is integrating NexusDesk and Class Notes Pipeline using the Lemma SDK.
- Lemma SDK will serve as the orchestrating layer, converting raw inputs into structured data and actionable tasks.
- The workflow involves Transcription and Syllabus Parsing Agents feeding into a central Learning & Execution Agent.
- Key build step: Migrating existing shell scripts to Lemma SDK Workflows for unified management.

Conclusion & Recommendations

The Gappy AI Hackathon Battle Plan presents a robust and highly promising strategy centered on the intelligent integration of two powerful, pre-existing tools: NexusDesk and the Class Notes Pipeline. By leveraging the Lemma SDK as the central orchestration layer, the team is set to create a sophisticated academic management system that not only captures and transcribes class content but also integrates it directly into a student's daily planner and project tracker. This approach provides a significant competitive edge by starting with mature, battle-tested components, thereby shifting the focus from foundational development to advanced integration and intelligent workflow design.

The proposed architecture, involving dedicated agents for transcription and syllabus parsing feeding into a core learning and execution agent, ensures a holistic and automated approach to academic data management. Recommendations for immediate action include finalizing the detailed specifications for each Lemma SDK Workflow, establishing clear API contracts between the agents and the NexusDesk frontend, and conducting thorough testing of each integrated module as it comes online. This strategic foresight and the strength of the existing components position the team for a high probability of winning the Gappy AI Hackathon, delivering a valuable and innovative solution for student productivity.